CODE ALPHA

TASK 1: Web Scraping ● Use Python libraries like BeautifulSoup or Scrapy to extract data from websites. ● Identify and collect relevant datasets from public web pages. ● If you don't code, use automated tools like Octoparse or ParseHub. ● Learn to handle HTML structure and web navigation to gather accurate data. ● Create custom datasets tailored to specific analysis needs.

```
!pip install requests beautifulsoup4 pandas

import requests

from bs4 import BeautifulSoup

import pandas as pd


# Website URL

url = "http://quotes.toscrape.com/"


# Send request

response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")


# Lists to store data

quotes = []

authors = []

tags = []


# Extract data

for item in soup.find_all("div", class_="quote"):

    quote = item.find("span", class_="text").text

    author = item.find("small", class_="author").text

    tag_list = [tag.text for tag in item.find_all("a", class_="tag")]


    quotes.append(quote)

    authors.append(author)

    tags.append(", ".join(tag_list))
```

```
# Create DataFrame
df = pd.DataFrame({
    "Quote": quotes,
"Author": authors,
    "Tags": tags
})

# Save to CSV
df.to_csv("quotes_data.csv", index=False)

print("Data saved successfully!")
print(df.head())
```

output:

```
Data saved successfully!
                                                Quote           Author  \
0  “The world as we have created it is a process ...  Albert Einstein
1  “It is our choices, Harry, that show what we t...     J.K. Rowling
2  “There are only two ways to live your life. On...  Albert Einstein
3  “The person, be it gentleman or lady, who has ...      Jane Austen
4  “Imperfection is beauty, madness is genius and...   Marilyn Monroe

                                           Tags
0       change, deep-thoughts, thinking, world
1                           abilities, choices
2  inspirational, life, live, miracle, miracles
3             aliteracy, books, classic, humor
4                   be-yourself, inspirational
```

#multi page scrapping

```
import requests
from bs4 import BeautifulSoup
import pandas as pd

base_url = "http://quotes.toscrape.com/page/{}/"

quotes = []
```

```
authors = []

for page in range(1, 11):   # Website has 10 pages
    url = base_url.format(page)
    response = requests.get(url)
    soup = BeautifulSoup(response.text, "html.parser")

    for item in soup.find_all("div", class_="quote"):
        quotes.append(item.find("span", class_="text").text)
        authors.append(item.find("small", class_="author").text)

df = pd.DataFrame({
    "Quote": quotes,
    "Author": authors
})

df.to_csv("all_quotes.csv", index=False)

print("All pages scraped successfully!")
print("Total Quotes:", len(df))
```

Output:

```
All pages scraped successfully!
Total Quotes: 100
```

```
#scrapping table data
import requests
from bs4 import BeautifulSoup
import pandas as pd

# Target URL
```

```
url =
"https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_populat
ion"

# Add a User-Agent header to avoid being blocked
headers = {"User-Agent": "Mozilla/5.0"}
response = requests.get(url, headers=headers)

# Parse the HTML
soup = BeautifulSoup(response.text, "html.parser")

# Find all tables with class "wikitable"
tables = soup.find_all("table", class_="wikitable")

print(f"Found {len(tables)} tables on the page.")

# Loop through each table and save to CSV
for i, table in enumerate(tables, start=1):
    rows = table.find_all("tr")
    data = []

    # Extract header row
    headers_row = [th.get_text(strip=True) for th in rows[0].find_all(["th", "td"])]

    # Extract data rows
    for row in rows[1:]:
        cols = [col.get_text(strip=True) for col in row.find_all(["td", "th"])]
        if cols:
```

```
        data.append(cols)

    # Create DataFrame
    df = pd.DataFrame(data, columns=headers_row)

    # Save each table to a separate CSV
    filename = f"wikitable_{i}.csv"
    df.to_csv(filename, index=False)

    print(f"Saved table {i} to {filename}")
    print(df.head(), "\n")

print("All Wikipedia tables scraped and saved successfully!")
```

output:

```
Found 1 tables on the page.
Saved table 1 to wikitable_1.csv
        Location     Population % ofworld         Date  \
0          World  8,232,000,000      100%  13 Jun 2025
1          India  1,417,492,000     17.2%   1 Jul 2025
2          China  1,404,890,000     17.1%  31 Dec 2025
3  United States    341,784,857      4.2%   1 Jul 2025
4      Indonesia    284,438,782      3.5%  30 Jun 2025

  Source (official or fromtheUnited Nations) Notes
0                        UN projection[1][3]
1                     Official projection[4]   [b]
2                       Official estimate[5]   [c]
3                       Official estimate[6]   [d]
4              National annual projection[7]

✅ All Wikipedia tables scraped and saved successfully!
```

#second code:

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import time
```

```
#  Define URL

base_url = "http://quotes.toscrape.com/page/{}/"


# Define Headers (Real Browser Header)

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                  "AppleWebKit/537.36 (KHTML, like Gecko) "
                  "Chrome/120.0.0.0 Safari/537.36",
    "Accept-Language": "en-US,en;q=0.9"
}


#Create Session

session = requests.Session()
session.headers.update(headers)
quotes = []
authors = []
tags = []


for page in range(1, 6):   # Scrape first 5 pages
    url = base_url.format(page)

    try:
```

```
        response = session.get(url, timeout=10)

        # Check status code
        if response.status_code == 200:
            print(f"Page {page} loaded successfully")
        else:
            print(f"Failed to load page {page} | Status: {response.status_code}")
            continue

        soup = BeautifulSoup(response.text, "html.parser")

        for item in soup.find_all("div", class_="quote"):
            quote = item.find("span", class_="text").text
            author = item.find("small", class_="author").text
            tag_list = [tag.text for tag in item.find_all("a", class_="tag")]

            quotes.append(quote)
            authors.append(author)
            tags.append(", ".join(tag_list))

        time.sleep(2)   # Delay to avoid blocking
         except requests.exceptions.RequestException as e:
        print("Error:", e)


df = pd.DataFrame({
    "Quote": quotes,
    "Author": authors,
    "Tags": tags
})
```

```
df.to_csv("complete_quotes_dataset.csv", index=False)


print("\nScraping Completed Successfully!")

print("Total Records:", len(df))

print(df.head())
```

output:

```
Page 1 loaded successfully
Page 2 loaded successfully
Page 3 loaded successfully
Page 4 loaded successfully
Page 5 loaded successfully

Scraping Completed Successfully!
Total Records: 50
                                                Quote           Author  \
0  “The world as we have created it is a process ...  Albert Einstein
1  “It is our choices, Harry, that show what we t...     J.K. Rowling
2  “There are only two ways to live your life. On...  Albert Einstein
3  “The person, be it gentleman or lady, who has ...      Jane Austen
4  “Imperfection is beauty, madness is genius and...   Marilyn Monroe

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